aiven EXOSCALE

INSIDE OPENSEARCH'S AI TOOLKIT

// FROM VECTOR SEARCH TO AGENTS

Dima Kan · OpenSearch Project user group, Vienna

// ABOUT THE SPEAKER

WHO IS TALKING



Dima Kan

Leading Search Product · AIVEN

Vector Podcast

- Managed OpenSearch at Aiven
- Contributed to Quepid, Luke, Lucene & Solr
- TomTom (planet-scale map search), AlphaSense (full document search) and Silo AI / AMD (web scale vertical search)

AIVEN PLATFORM

UNIFIED PLATFORM

STREAM

- Apache Kafka®
- Apache Kafka® Connect
- Apache Kafka® MirrorMaker 2
- Karapace
- Klaw

DATABASES

- PostgreSQL®
- ClickHouse®
- MySQL
- OpenSearch®
- Valkey
- Metrics
- DataHub®
- Grafana®
- Apps®

DEPLOY

aws · Google Cloud · Microsoft Azure · ORACLE Cloud · DigitalOcean · UpCloud · BYOC

TOOLING

Terraform · kubernetes · aiven CONSOLE · CLI | API

THIRD-PARTY INTEGRATION

Datadog · Prometheus · Amazon CloudWatch · GCP Monitoring · MongoDB · AWS S3 · GCP BigQuery · Couchbase · Snowflake · Splunk · Sumologic · Debezium · Google Pub/Sub · GCS

AGENDA

01. **The map**

Four layers of AI in OpenSearch, and where ML Commons sits.

02. **Search that understands**

Keyword, vector, hybrid and RAG. Live demo.

03. **Agents**

Tools, memory and multi-turn chat. Live demo.

04. **Quality + production**

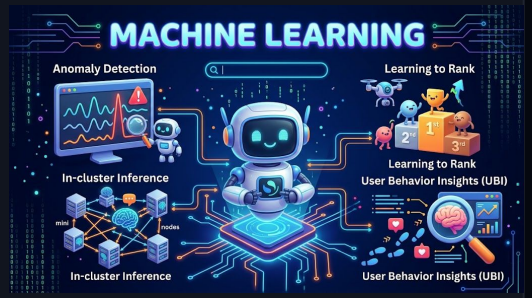
Measuring relevance and running it at scale.

THE MAP OF THE TOOLKIT

// THE TOOLKIT

FOUR LAYERS OF AI IN OPENSEARCH

Machine learning



Anomaly detection, in-cluster inference, learning to rank + user behavior insights (UBI).

AI-powered search



Keyword, vector and hybrid search, plus RAG pipelines built inside OpenSearch.

Agents



Orchestration, tools, memory and multi-turn conversations.

Search quality



The relevance workbench, LLM judges and agent-based tuning.

// ARCHITECTURE

ML COMMONS IS THE MODEL + AGENT REGISTRY

Neural + hybrid search
Embeds the query

Search pipelines
Rerank, RAG

Ingest pipelines
Embeds at index time

Agents + chat
Tools and memory

ML COMMONS PLUGIN · CALLS REFERENCE A MODEL ID

Model group
Access control

Model
Local or remote

Connector
Endpoint, auth, mapping

Agent
LLM plans and picks tools

Tools
Search, vector lookup, MCP

Memory
Chat history, long-term

In-cluster ML or data nodes
Pretrained models, CPU

Remote endpoints
Bedrock, OpenAI, SageMaker

// NOT EVERYTHING IS ML COMMONS

WHERE EACH FEATURE LIVES

FEATURE	PLUGIN	USES ML COMMONS?
Neural, hybrid + RAG search	neural-search, k-NN	Yes, for embeddings + LLMs
Agents + memory	ml-commons	It is ML Commons
Anomaly detection	anomaly-detection	No, own algorithms
Learning to rank	learning-to-rank	No, own model store
User behavior insights	ubi	No, collects events
Search relevance workbench	search-relevance	Yes, LLM judges via connectors

ML COMMONS UNDER THE HOOD

// ML COMMONS

FOUR PRIMITIVES TO KNOW

1 **Model group**
A named container with access control. Models are versions inside it.

2 **Model**
A pretrained model in the cluster, or a remote model. Everything else refers to it by ID.

3 **Connector**
The endpoint, credentials and request mapping for a remote model such as an LLM.

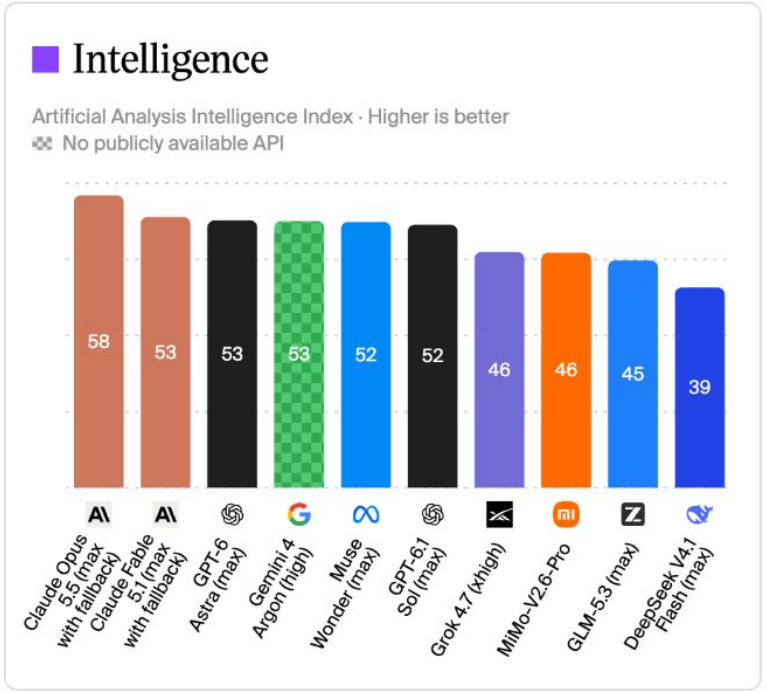
4 **Task**
Register and deploy run asynchronously. Poll the task until it completes.

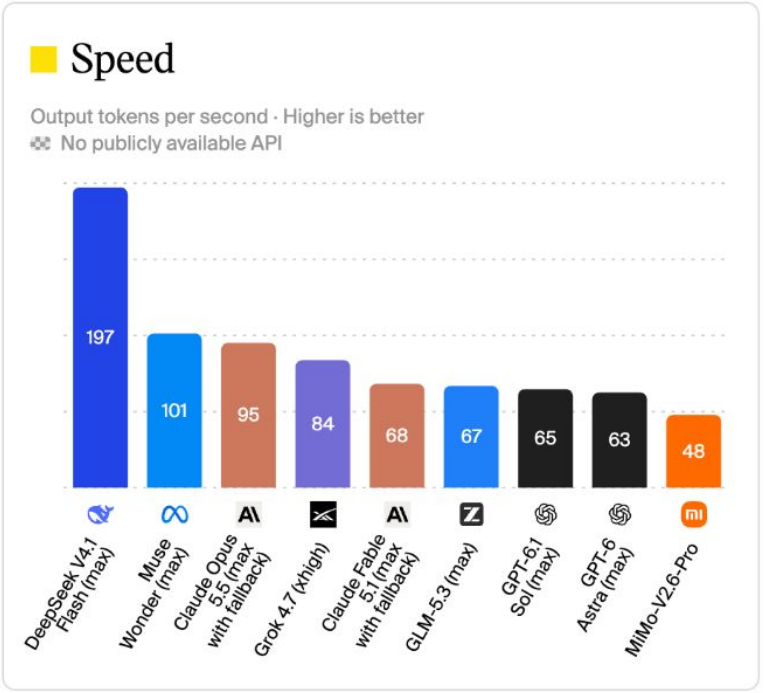
// CHOOSING A MODEL

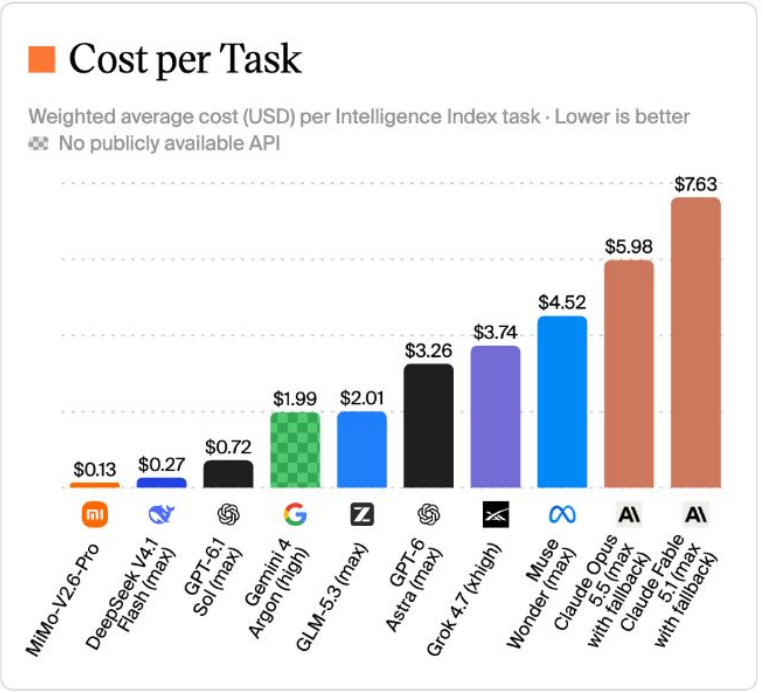
LOCAL OR REMOTE MODEL?

QUESTION	IN-CLUSTER PRETRAINED	REMOTE CONNECTOR
Where it runs	On your nodes, CPU	At the provider
Setup	Register, then deploy	Allow endpoint, connector, register
Cost driver	Cluster CPU + RAM	Tokens or API calls
Data leaves cluster	No	Yes, to the provider
Good fit	Embeddings, small models	LLMs, large embedding models

// CHOOSING A MODEL







https://artificialanalysis.ai/

// AI-POWERED SEARCH

KEYWORD SEARCH FINDS THE WORDS YOU TYPED. VECTOR SEARCH FINDS WHAT YOU MEANT. HYBRID DOES BOTH.

Aa

Keyword (BM25)

Exact terms, IDs and error codes. Fast, with scores you can explain.

◇

Vector

Meaning, synonyms and paraphrase. Needs an embedding model.

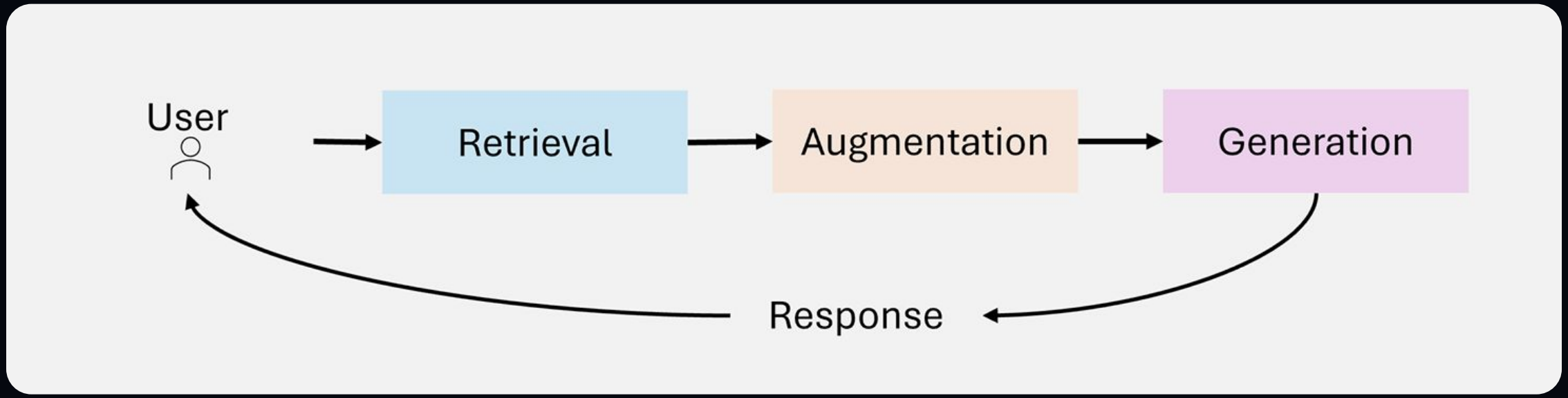
+

Hybrid

Both scores are normalized and blended by a search pipeline.

// AI-POWERED SEARCH

RAG IN A NUTSHELL



- Published Apr, 2021 by Patrick Lewis (co:here) et al (FAIR = Facebook AI Research)
- Was an “accidental” find by a team at Facebook
- Adds domain and topical grounding to an LLM
- https://arxiv.org/pdf/2005.11401

// AI-POWERED SEARCH

RAG VS FINE-TUNING

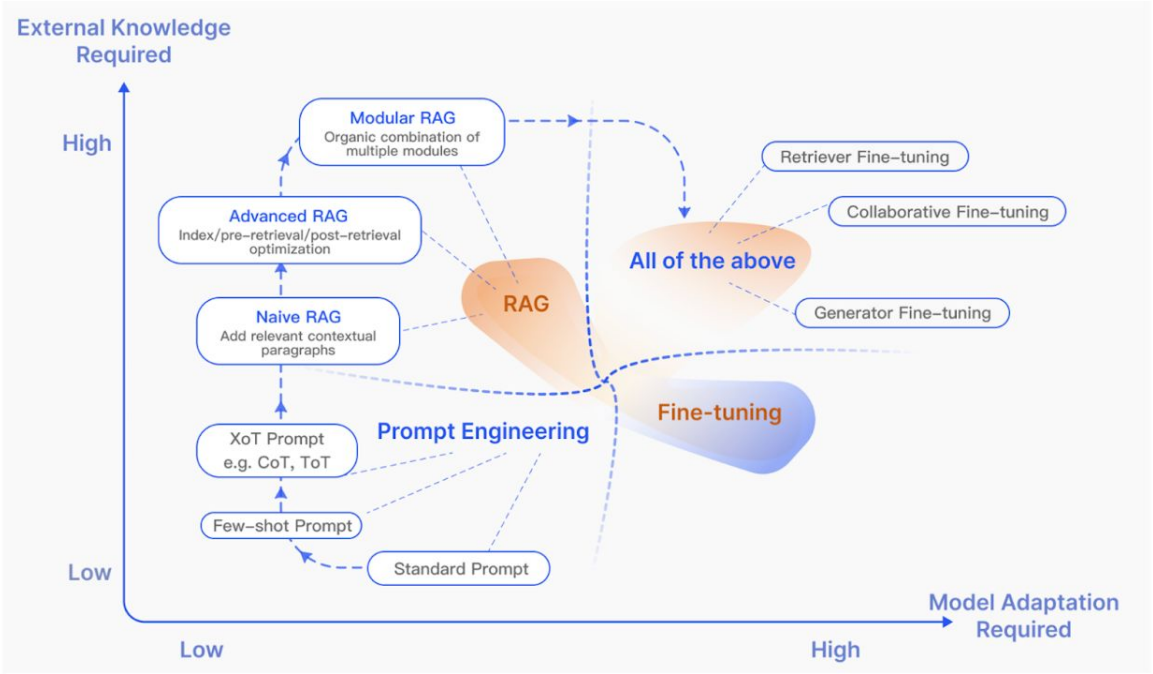
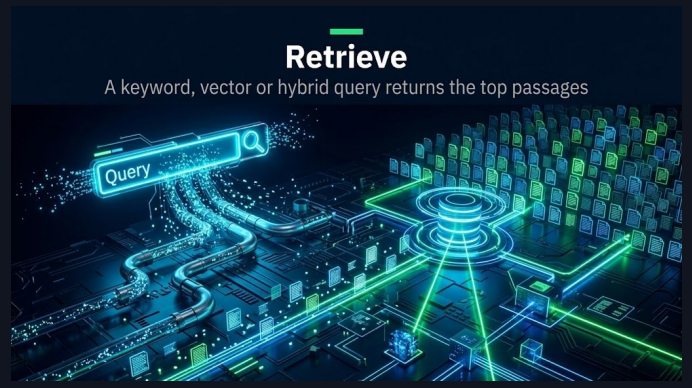


Fig. 4. RAG compared with other model optimization methods in the aspects of "External Knowledge Required" and "Model Adaption Required". Prompt Engineering requires low modifications to the model and external knowledge, focusing on harnessing the capabilities of LLMs themselves. Fine-tuning, on the other hand, involves further training the model. In the early stages of RAG (Naive RAG), there is a low demand for model modifications. As research progresses, Modular RAG has become more integrated with fine-tuning techniques.

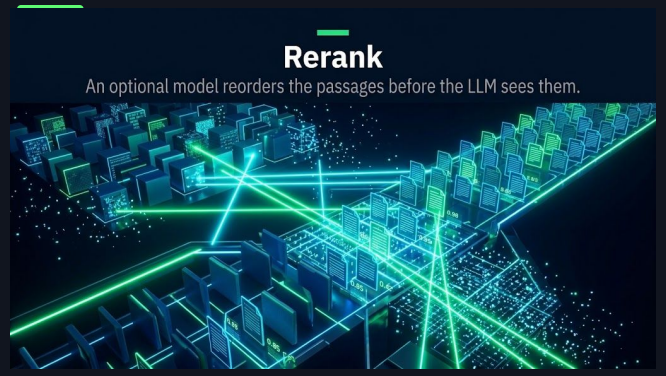
- Shift to Agentic RAG
- Paradigm shift from 'static' RAG systems by introducing autonomous agents
- Agents:
  - dynamic decision-making
  - Workflow optimization
- Iterative refinement
- Adaptive retrieval
- Address complex, real-time, multi-domain queries

// RAG

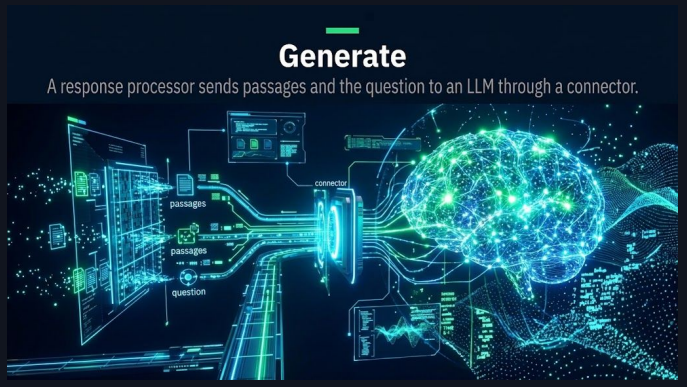
RAG IS A SEARCH PIPELINE



Many issues of RAG are in the retrieve phase



Advanced RAGs will reason about the response and if need be, will query for more data to achieve the original goal



During generation you can tune the LLM or pick the LLM that is most suitable for this task

// LIVE DEMO 1

THE EMBEDDING LOOP

- Register + deploy a pretrained model
- Embed documents at ingest
- Compare keyword, neural and hybrid
- One model ID, three features

```
# Dev Tools
POST /_plugins/_ml/models/_register
POST /_plugins/_ml/models/<id>/_deploy
PUT  /_ingest/pipeline/embed-pipeline
GET  /talk-docs/_search
GET  /talk-docs/_search?search_pipeline=hybrid-pipeline
```

Advanced configuration

Advanced parameters to configure your service. Learn more

There's no way to reset advanced configuration options to default. Options that you add cannot be removed later. Changing or adding some options causes the service to restart.

Add configuration options

opensearch.ml_commons_only_run_on_ml_node In sync

Enable or disable running ML Commons tasks only on ML nodes. When enabled, ML tasks will only execute on nodes designated as ML nodes. Defaults to true.

Disabled

opensearch.ml_commons_trusted_connector_endpoints_regex In sync

Adds the trusted endpoints to the cluster settings. Supports Java regex expressions.

^https://ai\.aiven\.io/v1.*$

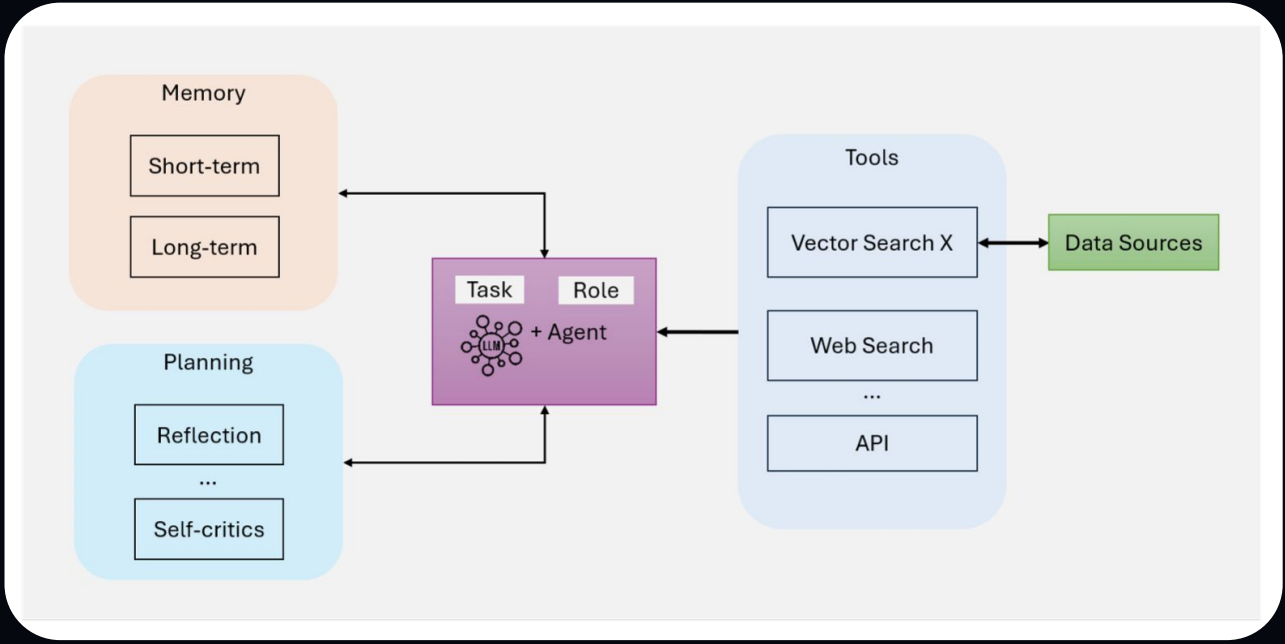
Cancel Save configuration

AGENTS THAT USE TOOLS

// AGENTS THAT USE TOOLS

AI AGENTS: COMPONENTS

- **LLM** (with defined role and task)
- **Memory**
  - Short-term tracks immediate conversation state
  - Long-term: accumulated knowledge and experiences
- **Planning**
  - Reflection, query routing, self-critique
- **Tools**
  - Vector / web search, APIs



// AGENTS THAT USE TOOLS

LLM. SYSTEM PROMPT FOR THE AGENT.

System prompt (role+task) for the agent:

"You are an AI Research Librarian helping students in an LLM/RAG course.
Your job:

- Understand the student's research question.
- Search their local paper library and the web for relevant sources.
- Summarize and explain papers in simple language.
- Keep track of what the student has already read and their preferences.

Always:

- Show which sources you used.
- Admit when you're unsure and suggest how to verify."

// AGENTS THAT USE TOOLS

MEMORY (SHORT-TERM)

User: “I want to build a RAG system for legal contracts. What should I read first?”

Agent (internally tracks in **short-term** memory):

- Current topic: **RAG for legal contracts**
- User level: says they’re in an **LLM/RAG course**, so likely intermediate
- Last question: **“what should I read first?”** (guides next response)

json

```
{
  "current_topic": "RAG for legal contracts",
  "user_level": "intermediate student",
  "last_question": "what should I read first?"
}
```

// AGENTS THAT USE TOOLS

MEMORY (LONG-TERM)

- Accumulates over time, over multiple interactions with the user
- Won’t recommend the papers the user has already read for their question: “How do I evaluate my RAG system?”

json

```
{
  "user_profile": {
    "name": "N/A or anonymized",
    "background": "CS undergrad, some Python, new to RAG",
    "interests": ["RAG", "legal domain", "evaluation"],
    "reading_history": [
      "Lewis et al. 2020 - Retrieval-Augmented Generation",
      "Self-RAG 2023",
      "GraphRAG blog (Microsoft, 2025)"
    ],
    "preferences": {
      "wants_simple_explanations": true,
      "likes_concrete examples": true
    }
  }
}
```

// AGENTS

FOUR AGENT TYPES

Flow

Runs tools in a fixed order. No LLM required.

Conversational flow

Fixed tool order, with chat memory.

Conversational

The LLM chooses tools and loops until it can answer.

Plan, execute, reflect

A planner breaks down hard questions and checks the results.

// AGENT BUILDING BLOCKS

TOOLS, MEMORY + MCP

Tools

Built-in tools such as search and vector lookup, plus your own.

Memory

Conversation history today. Agentic memory adds long-term facts, with retention policies in 3.8 (experimental).

MCP

3.8 extends the Model Context Protocol to all four agent types, so agents can call external tool servers.

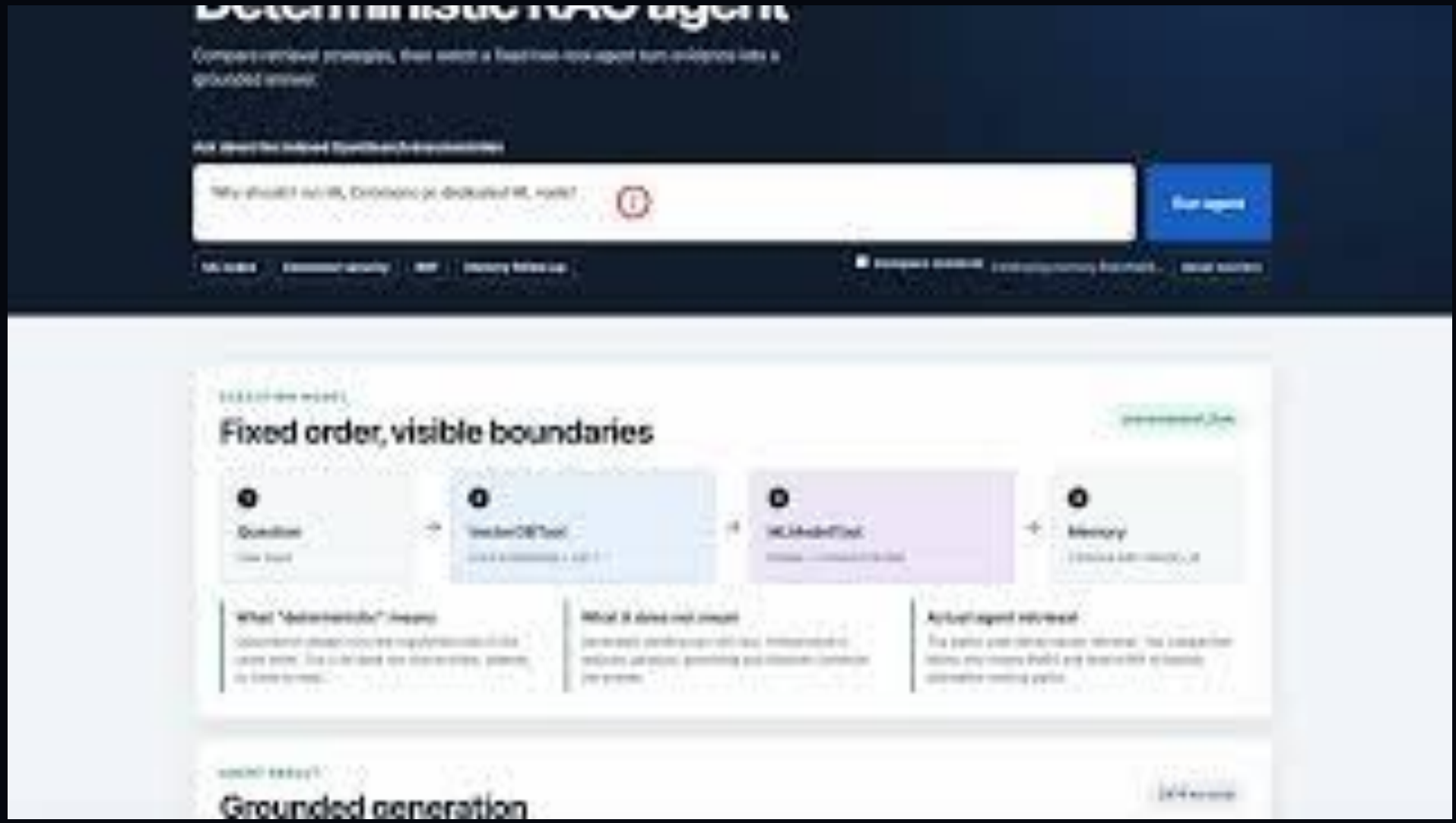
// LIVE DEMO 2

FROM A RECIPE TO AN AGENT

- A flow agent: one search tool, no LLM
- A conversational agent: the LLM picks the tool
- A follow-up that only works with memory
- Read the agent's trace

```
# Dev Tools
POST /_plugins/_ml/agents/_register  flow
POST /_plugins/_ml/agents/<id>/_execute
POST /_plugins/_ml/agents/_register  conversational
POST /_plugins/_ml/agents/<id>/_execute  memory_id
GET  /_plugins/_ml/memory/<id>/messages
```

// RECORDED DEMO 2



// SEARCH QUALITY

RELEVANCE NEEDS A FEEDBACK LOOP

Search relevance workbench

Compare query setups side by side on a set of queries.

LLM as a judge

In 3.8 any ML Commons connector can produce relevance judgments.

User behavior insights

UBI collects queries and clicks, so judgments reflect real use.

OpenSearch Dashboards

Search Relevance

DATA SOURCE Local cluster

a

Workbench

Experiments
- Query Analysis
- Query Set Comparison
- Search Evaluation
- Hybrid Optimizer

Query Sets

Search Configurations

Judgments

Search

Search

Setup 1

Data Source: Local cluster

Index: talk-docs

Search Configuration:

Pipeline - *optional*:

Query Help

```
{
  "query": {
    "match": {
      "text": "ML"
    }
  }
}
```

Enter a query in OpenSearch Query DSL. Use %SearchText% to refer to the text in the search bar

Setup 2

Data Source: Local cluster

Index: talk-docs

Search Configuration:

Pipeline - *optional*:

Query Help

```
{
  "query": {
    "match": {
      "text": "Commons"
    }
  }
}
```

Enter a query in OpenSearch Query DSL. Use %SearchText% to refer to the text in the search bar

Results for query:

Display Field: Text

Size: Medium



Setup 1 Results
(4 results)

1. By default, ML Commons runs local models and ML tasks only on nodes with the ml role. Dedicated ML nodes isolate inference CPU and native-memory use from indexing and search.
2. ML Commons defaults to at most 10 deployed models and 10 running ML tasks per node. Its circuit breakers default to 90 percent native-memory usage, 85 percent JVM-heap usage,
3. Remote model credentials belong in the connector credential object, not in each prediction request. ML Commons encrypts connector credentials with AES GCM and
4. A local model is first registered and then deployed. Both operations are asynchronous and return ML task IDs that can be polled. Registration stores model metadata and artifact

Setup 2 Results
(3 results)

1. Remote model credentials belong in the connector credential object, not in each prediction request. ML Commons encrypts connector credentials with AES GCM and…
2. By default, ML Commons runs local models and ML tasks only on nodes with the ml role. Dedicated ML nodes isolate inference CPU and native-memory use from indexing and search.…
3. ML Commons defaults to at most 10 deployed models and 10 running ML tasks per node. Its circuit breakers default to 90 percent native-memory usage, 85 percent JVM-heap usage,…

// TUNING

THE TUNING LOOP

1 **Collect**
Capture queries and clicks with UBI.

2 **Judge**
Rate results with people, click data or an LLM.

3 **Experiment**
Run query variants in the workbench and compare metrics. An agent can suggest variants.

4 **Ship**
Promote the winner, then repeat as content and user behavior change.

// PRODUCTION

TRADE-OFFS AT SCALE

DECISION	CHEAPER	FASTER OR BETTER	WATCH
Embedding model	Small, in-cluster	Larger, remote	Remote adds latency + per-call cost
Where inference runs	Shared data nodes	Dedicated ML nodes	Inference competes with search
Vector memory	Compressed vectors	Full precision	Recall vs RAM, test on your data
LLM calls	Small model, fewer calls	Large model every time	Tokens drive the bill
Streaming	REST	gRPC (new in 3.8)	Lower latency, more setup

// ON AIVEN FOR OPENSEARCH

WHAT TO KNOW BEFORE YOU RUN IT

1

CPU only on Aiven

On Aiven, ML tasks run on CPU today. GPU-backed ML nodes are planned.

2

Choose where ML runs

Dedicated ml nodes on eligible plans. Otherwise set ml_commons_only_run_on_ml_node to false.

3

Pretrained or remote

No custom model upload. Deploy a pretrained model yourself, or add a connector.

4

Allow endpoints, restrict access

Add providers to the trusted endpoint regex. Enable OpenSearch Security to limit who uses models.

// TAKEAWAYS

FOUR THINGS TO REMEMBER

1

A model is an ID

Register once. Ingest, search and agents just reference it.

2

Start with hybrid search

Keyword for exact terms, vectors for meaning, one pipeline to blend them.

3

An agent is a recipe of tools

Begin with a flow agent, add an LLM and memory when you need them.

4

Measure before you tune

Judgments and click data beat intuition.

YOUR FREE €50 COUPON QR CODE.

Exoscale is offering €50 coupon codes to explore OpenSearch.





Dmitry Kan

Head of Search Product
Host of @Vector Podcast

// THANK YOU

QUESTIONS?

dima.kan@aiven.io · aiven.io

References

- Demo code: https://github.com/Aiven-Labs/opensearch-ml-commons-demo
- ML with OpenSearch: https://docs.opensearch.org/latest/ml-commons-plugin/
- Stop Calling an Embedding API: Run the Model Inside Opensearch https://www.linkedin.com/pulse/stop-calling-embedding-api-run-model-inside-octavian-balaban-okvje/
- Building agents that remember: https://aiven.io/blog/building-agents-that-remember-the-opensearch-developer-tier
- My recent blog on UBI (User Behaviour Insights): https://aiven.io/blog/your-users-already-know-whats-relevant-are-you-listening
- Going beyond hybrid search: https://aiven.io/blog/beyond-hybrid-search-traversing-vector-spaces-with-wormhole-vectors
- https://aiven.io/opensearch
- Vector Podcast in different form factors: https://www.vectorpodcast.com/
- Connect with me on LinkedIn: https://www.linkedin.com/in/dmitrykan/